AsiRo-μ: A Multi-purpose Robotic Assistant for Educational Inclusion of Children with Multiple Disabilities

Jonnathan Andrade-Altamirano[1], Ana Parra-Astudillo[1], Vladimir Robles-Bykbaev[1(✉)], Nidia Almeida-Solíz[1,2], Sofía Bravo-Buri[1], and Efren Lema-Condo[1]

[1] Universidad Politécnica Salesiana, Cuenca, Ecuador
jandradea5@est.ups.edu.ec, {vrobles,aparraa,nalmeida, elema}@ups.edu.ec
[2] Unidad Educativa Eugenio Espejo, Cuenca, Ecuador

Abstract. Nowadays, artificial intelligence is a tool widely used in several areas such as medicine, personal assistance, regular/special education, leisure, among many others. Additive manufacturing development allows designing and making innovative low-cost robotic assistants for educational inclusion processes in developing countries. There are not enough assistive technologies for special education centers in Ecuador due to the lack of resources. For these reasons, in this paper, we describe the robot AsiRo-μ (multi-purpose robotic assistant, for its Spanish name). AsiRo-μ is based on the open-source 3D printed robot Inmmov. It implements the following functionalities: hand gestures recognition, automatic speech recognition and text to speech function through the IBM Watson Cloud services, and gesture imitation. The robot aims to conduct a dialogue with the children, motivate them to carry out the exercises/rehabilitation activities, and motivate and engage them in the therapy sessions (for children with disabilities). To determine the children's acceptance level, we carried out a pilot experiment with two groups of children. The first group consisted of 7 children with multiple disabilities (intellectual disability, Joubert syndrome, Down Syndrome, Autism, etc.), whereas in the second participated 30 children aged between 4 and 6 years (without disabilities). Each child interacted with the robot, and a group of experts evaluated the children's perceptions.

Keywords: Robotic assistant · Equalization · Microcontroller · Image processing · Segmentation · Computer Vision

1 Introduction

Talking about education is exciting, describing learning theories, sharing experiences on didactic processes or methodological strategies applied in the classroom, enriches the teaching work, even more so when there is commitment, vocation and love for teaching. However, the least addressed topic in these meetings is the applicability of such strategies

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with students who have Special Educational Needs (SEN); educational inclusion still generates some fear, perhaps due to the lack of training received as teachers, the lack of resources or strategies to be used or simply because of the social prejudices that exist.

Knowing each student allows proposing plans that contain strategies that promote equal opportunities based on their individual needs and interests, improving the classroom environment, providing an active role to each student in order to optimize their learning and achieve true educational inclusion.

The Ministry of Education, in the Pedagogical Strategies Work Guide for Special Educational Needs, proposes intervention mechanisms based on diversified strategies such as the setting and adaptation of the classroom, maintaining channels of communication with families and therapists.

The methodology that rules the teaching-learning process at the levels of Initial Education and High School is the game, which is a strategy that allows the integration of students with and without SEN associated or not with disabilities. At these levels of education, inclusion processes are carried out in a very natural way, since in early childhood, children accept their peers without prejudice and their characteristic of solidarity favors the support in the acquisition of skills according to their abilities. When continuing in the Basic Education levels, these processes are broken and the methodologies used vary according to the strategies that each teacher considers optimal for the development of their classes [1].

It is at these levels where the continuity of inclusive strategies should be proposed, through collaborative work that facilitates the management of various levels within the classroom, each student has an active role: investigating and experimenting with the guidance of their teacher who facilitates and raises situations or topics that help them to discover and learn in an active way.

Therefore, it is important to highlight the importance of continuous training in inclusion and new methodological strategies that allow the development of meaningful learning.

2 Related Work

This section the importance of the use of robotic assistants and inclusive software developed to support children of different ages and with different needs.

It has been shown that the work of children with Down Syndrome with humanoid robots is very important for their educational development and social skills, is what Lehmann et al. proved, by conducting an experiment with two robots, the first robot has a humanoid form (Kaspar) and has the size of a child, while the second has the shape of a small car, the experiment was conducted for three months with a girl with Down Syndrome; one of the activities that was performed with the girl was the ability to mimic a complete sequence of movements. After 3 months it was concluded that the girl preferred to work with the humanoid-shaped robot [2].

As autism is a neurological disorder that manifests itself in the first years of life and alters the normal functioning of the brain, it causes some inconveniences in children such as not being able to interact and communicate easily with the environment around them, pronouncing unintelligible words in a low voice, responding incorrectly to any

question and not presenting fluency in their dialogue, these drawbacks were obtained in the research conducted by [3], developed for three months with a large population of children with autism between 4 to 11 years old.

With the results obtained from this research, a computer game has been developed, which is based on the use of pictograms that appear randomly on the screen, the children must pronounce the name of each object and then form sentences, which has caused children with autism to be able to have a fluent dialogue [3].

Likewise, an interdisciplinary team developed "*An intelligent system based on ontologies and ICT tools to support the diagnosis and intervention of children with autism*", designed to improve the social skills of people with autism, which adapts to the needs and abilities of each user in order to provide a personalized therapy. For this research a robotic assistant was built in the shape of a stuffed animal and was evaluated with 30 children who expressed their agreement with its appearance, movements and functionalities [4].

3 Proposed Method

3.1 Robotic Assistant

There is a need to develop a low-cost robotic assistant to interact with children with and without disabilities, capable of perform voice recognition and sign recognition using artificial intelligence techniques and with a friendly design.

Figure 1 shows the architecture diagram of the robotic assistant proposed in this article. Therapists, teachers, and psychologists are represented at the top of the figure, who will be in charge of working with the robotic assistant in their classrooms and in their different work sessions.

The end users and the visual, auditory, motor and intellectual stimuli that the tool will allow to develop with the use of the proposed modules are shown below.

The implementation of the robotic assistant was based on the INMOOV animatronic model, its website presents open access resources (http://inmoov.fr/), further, an additive manufacturing through 3D printing was used for the construction of the head and all its internal parts, whose links will move thanks to the actuators that form the head.

The robotic assistant has four degrees of freedom distributed in the eyes, jaw and trachea, for which the tool is composed of the electronic components shown in Fig. 2, the motion control of the actuators is performed with the Arduino MEGA 2560 microcontroller together with a 16-channel, 12-bit PWM controller PCA9685.

The actuators used for the movement of the joints are detailed below. Two DS929V micro-Servos are used for the movement of the eye mechanisms, a Metal Gear MG995 Servo is used for the movement of the jaw mechanism, and a Mega Giant HS-805BB Servo is used for the movement of the head mechanism.

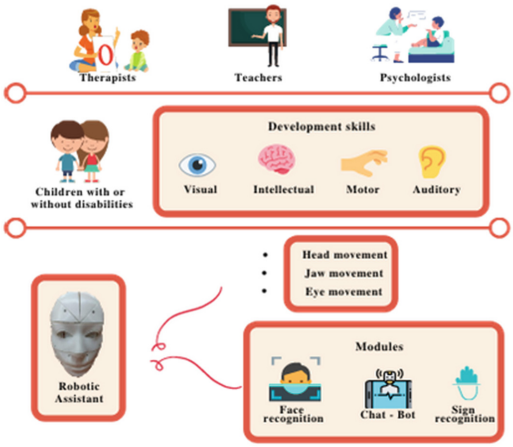


Fig. 1. System architecture diagram

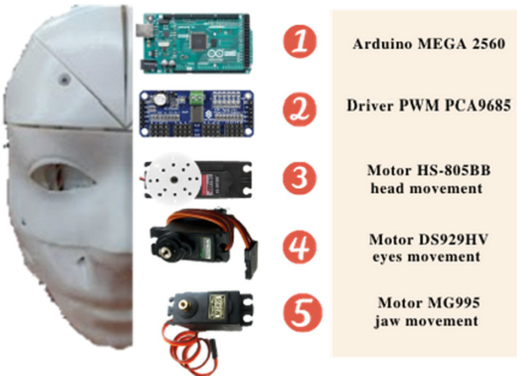


Fig. 2. General diagram of the robotic assistant, in the left section you can see its electronic components and in the right section you can see its physical aspect.

3.2 Gesture Recognition System

Displayed One of the main points for the robotic assistant to perform interaction is the artificial vision system, which consists of two parts: object detection and object identification. In order to perform the gesture detection process, first, it is necessary to work on the lighting of the scene, just as with the human eye, it is possible to have a better projection of the area that is being observed according to the amount of light that is available at that moment.

The histogram equalization technique has been used to improve the illumination, which alters each pixel of the image, returning a new image with a better contrast, to move on to a second point such as segmentation, which consists of taking a scene in the RGB color space (called like that because of the composition of its three channels corresponding to red, green and blue) and creating a binary image where the object is

separated from the rest of the scene based on its color intensity, being white the space occupied by the object and black the color of the rest of the scene, it is possible that after performing a segmentation there may be pixels left over that can be considered as noise, which can be eliminated by applying blur and threshold filters.

This process allows the detection of objects based on their color, in this case the pixels that represent the skin are taken. The second part is the application of object identification techniques, in this case the Sklearn MLPClassifier python library (Multi-layer Perceptron Classifier) has been used, algorithm that relies on an underlying neural network to perform classification tasks, which optimizes the loss function using stochastic gradient descent [5].

It has been proposed to differentiate a gesture to initialize the interaction with the robotic assistant, another gesture to identify a greeting, a gesture to move to the next question and a gesture to try again, 500 images for each gesture of its binary form (black and white) have been sampled to create the classifier, the samples have been taken from both the right and the left hand. The Hu invariant moments have been extracted from each sample, a technique first proposed by Ming-Kuei Hu in [6], which can recognize objects no matter how different their size, position and scale are, minimizing the loss of information, also considered as the weighted average of the pixels of an image.

For the construction of the network, the 7 invariant moments of Hu are used in the input layer, which represent each image with a gesture, in the binary color space, moreover, an element for the processing of the output layer is used, which indicates the number by which the gesture will be represented, as shown in Fig. 3. Although it is intended to identify only 4 gestures, the network has been trained with 7 different gestures, to avoid false positives with gestures that may appear similar in different positions or distances of the hand with respect to the camera, once the network is trained, it returns an accuracy of 0. 92% of effectiveness.

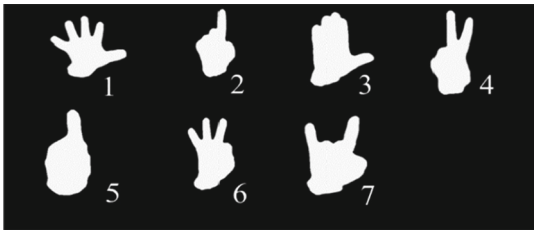


Fig. 3. Representation of the gestures considered to create the classifer

3.3 Systems Integration

The integration with the robotic assistant consists in that once the child puts his hand in front of the robot, the robot recognizes the gesture and asks a question, together with the jaw gesticulation to simulate the movement of speech, then he has a between 10 to 20 s to make a short recording depending on the case in which the child can answer, the audio file is captured and through the IBM Watson Speech To Text API the audio is processed

to convert it into text, this enters an algorithm where keywords are searched to validate that the answer is correct, if the answer is correct the robotic assistant will play an audio saying "correct, now you can perform a gesture to move to the next question", at that moment the gesture that corresponds to the next question must be done, and the robotic assistant will ask another question, in case the answers are considered incorrect the assistant will reproduce the following sentence "incorrect, could you make the gesture to try again", at that moment the child must make the gesture that corresponds to try again (Fig. 4).

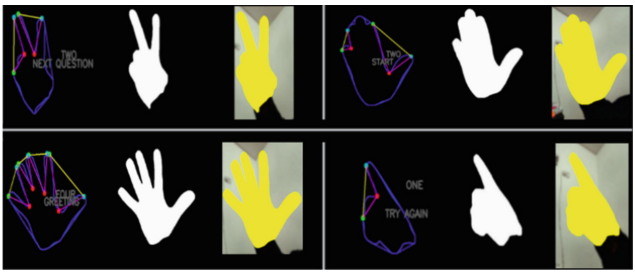


Fig. 4. Gestures used to perform the interaction with the robotic assistant.

4 Pilot Experiment and Preliminary Results

In order to determine the real feasibility of using the robotic assistant as an educational support tool, an experimentation plan was carried out with a group of 30 children without disabilities and 6 children with multiple disabilities. The first group of 30 children are students from the "Eugenio Espejo" Educational Unit, between the ages of 4 and 6 years (mean = 4.6, and SD = 0.844). A total of 18 boys and 12 girls participated in the experiment.

An interaction process was carried out to measure the children's perception of the robotic assistant AsiRo-μ, by using the assistant's gesture recognition and automatic speech recognition system. Then, they were asked to fill out a survey consisting of 13 questions related to the appearance and functionalities of the robot, as well as the possibility of having the assistant at home, and its fluency in speech and movements (mouth). Also, in order to protect the children's privacy, only 2 demographic questions (age and gender) were considered.

Each survey question was designed on a Likert scale where 1 represents "totally disagree" and 5 represents "totally agree". The survey was validated with the internal consistency model, Cronbach's alpha, obtaining a result of 0.86. Thus, it can be affirmed that the reliability used to measure the perception of the 30 children is appropriate.

As can be seen in Fig. 5 (top plot), 60% of the children consider that they "strongly agree" and "agree" that the size and shape of the robot are appropriate. Along the same line, 76% and 83% of the children liked the colors of the robot and the base that holds the robot's head, respectively. It is important to mention that the speed of the robot's

mouth movements should be improved, since 50% of the children consider them as "totally adequate" and "adequate". As for the robot's voice, 83% considered it as "very pleasant" and "pleasant".

On the other hand, the lower plot of Fig. 5 shows the children's perception of the robot's functionalities. 97% of the children consider the activities provided by the robot as "very interesting" and "interesting", while 70% think that they would "totally agree" and "agree" to have the robot at home. One aspect that stands out is that children prefer to talk to the robot (73%) compared to interacting through hand gestures (50%).

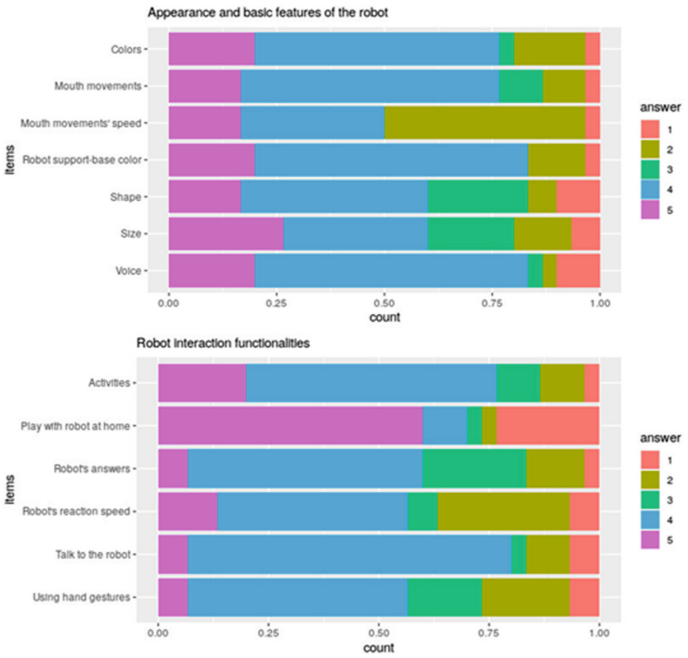


Fig. 5. Results of children's perception of the physical characteristics (top) and functionalities (bottom) of the robot.

On the other hand, we worked (virtually) with 7 children (2 girls and 5 boys) with disabilities (4 with Down Syndrome, 1 with intellectual disability and autism, 1 with deep bilateral anacusis, and 1 with chronic encephalopathy). A therapist guided each working session with the support of a team of engineers. We used the following criteria to determine a preliminary children's perception regarding the robotic assistant: a) understand how to interact with the robot, b) difficulty to perform the interaction (through voice commands and hand gestures), c) attention period, and d) what they think to have a robot at home. The average results (using a Likert scale where 5 is "totally agree" and 1 is "totally disagree") were 2.7, 2.28, 3, and 4.8, respectively.

5 Conclusions

Through the analysis conducted from the application of the surveys, it is concluded that the robotic assistant AsiRo-μ produced a great impact on children between 4 to 6 years old with and without disabilities, which can be verified with the value of Cronbach's coefficient, which is 0.86, showing that the survey reflects consistency between the items.

Likewise, the perception of the children with respect to the size, color, voice and shape of the robotic assistant has a positive acceptance, since it is in scales of "Totally agree" and "Strongly agree", "Very pleasant" and "Pleasant", in addition the results indicate that the children prefer to interact with the assistant verbally than using manual signs, as indicated in the section of the experimentation plan and preliminary results.

In the case of children with intellectual disabilities, it has been concluded that due to their difficulties in the motor area, it is difficult for them to imitate the requested gestures, they tried, but most of them required physical support from the person who accompanied them. In terms of listening to the assistant, it was easier for them to understand the question, although they were given verbal support by repeating the question. It is worth mentioning that the questions were repeated a second time and this time the children understood the instructions a little better, and even tried to make the gestures they liked or found easier on their own. This experience is encouraging because it opens the door to future improvements by adapting gestures and providing the opportunity to work with a robotic assistant during therapies.

Due to these results, it can be concluded that an interactive tool has been created, which will favor children in the first educational stage with or without SEN, in line with the purpose of the GIIATa Research Group and the UNESCO Chair on Support Technologies for Educational Inclusion of the Politecnica Salesiana University, Campus Cuenca.

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